



BEGAN

IT CAME FROM OUTER SPACE

It is possible that some of the chemicals that form the building blocks of life came from elsewhere in the galaxy. In its early days, Earth was under constant bombardment from comets, asteroids, and meteors. Scientists have discovered sugars and amino acids in meteorites. Both of these substances are used to make larger molecules called proteins, which build and maintain cells.



METEOR
SHOWER



DEEP SEA VENTS

Perhaps life began around openings in the ocean floor called hydrothermal vents. The heated water spewing out of these vents may have provided the energy for chemical reactions. Bacteria have been found living in these vents, surviving on the sulphurous compounds they emit, without the need for light or oxygen. These conditions are probably similar to those that existed in the early days of Earth.

STROMATOLITES



INDEPENDENT LIFE

The earliest physical evidence we have for the formation of cells comes from fossilized structures called stromatolites. These are around 3.5 billion years old, although scientists think the first cells may have originated 3.8 billion years ago. Stromatolites played a crucial role in oxygenating the atmosphere so life could move onto the land. Stromatolites are still forming today, and are made of mats of cyanobacteria (blue-green algae).